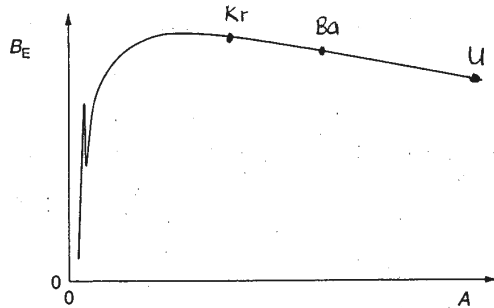


$$E \approx 1.29 \times 10^{-5} \text{ J}$$

5 (a) Maximum binding energy per nucleon $\approx 8.8 \text{ MeV}$

(b)



(c) The total binding energy of the products (Ba and Kr) is a larger value than the reactant (U). This means that more energy is released when the Ba and Kr nuclei are formed than the energy required to break up the U nucleus. Hence, the overall effect is that energy is released which resulted in more stable products being formed that have a higher binding energy per nucleon.

6 (a) (i) Gravitational potential at a point in the gravitational field is the work done per unit